

Math 170S - Homework 3 - due January 30

Problem 1. Let $X_1, \dots, X_n$ be the sample taken from uniform distribution on the interval $[-\theta, \theta]$. Find maximum likelihood estimator for the parameter θ .

Problem 2. For $i = 1, \dots, n$, let us model unknown $Y_i = \alpha + \beta x_i + \varepsilon_i$, where x_i 's are considered known, $\varepsilon_i \sim N(0, \sigma^2)$ and α, β, σ^2 are unknown parameters of the model.

- a) What is the likelihood function $L_{x_1, \dots, x_n, \alpha, \beta, \sigma^2}(y_1, \dots, y_n)$ for this model?
- b) Compute partial derivatives of the log-likelihood function and check that MLE estimates for the parameters are

$$\hat{\beta} = \frac{\sum_{i=1}^n y_i(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2},$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = \bar{y} - \frac{\sum_{i=1}^n y_i(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \bar{x},$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta}x_i)^2.$$

- c) Check that all 3 estimates above are unbiased.

Problem 3. 6.5-4

Problem 4. 6.7-2

Problem 5. Gamma distribution is defined by its density function

$$f(t; \theta, a) = \frac{\theta^a}{\Gamma(a)} t^{a-1} e^{-\theta t}$$

when $t \geq 0$ and 0 otherwise. Here, θ is unknown parameter, a is a known parameter $\Gamma(a)$ is a normalization constant.

- a) Check that Gamma distribution belongs to the exponential family of distributions.
- b) Use this fact to give a sufficient statistic for θ .
- c) Give examples of two more different sufficient statistics for θ .

Problem 6. a) Show that $\sum_{i=1}^n X_i$ and $\sum_{i=1}^n X_i^2$ are jointly sufficient statistics for two unknown parameters of the normal distribution $N(\theta_1, \theta_2)$ (based on the data sample $X_1, \dots, X_n$)

- b) Show that $\bar{X}$ and s^2 are jointly sufficient statistics for the same distribution.
- c) Give yet another example of a couple of jointly sufficient statistics.

Hint: Example 6.7-5 in Hogg et al. Anyway, make sure to include detailed solution into your homework (not just refer to the aforementioned example). If you claim that something is a sufficient statistic as a function of another sufficient statistic, have you checked that this function is invertible?